

Warm-Glow Bequests and Descendant Risk

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Abstract

Reduced-form bequest motives are routinely assumed to exhibit decreasing absolute risk aversion (DARA). I show this is consistent with an underlying altruistic primitive: when descendant risk is unrealised at the bequest choice date, DARA of the warm-glow preference is inherited from the descendant's continuation value under both multiplicative and additive risk, without parametric restriction. Yet the two canonical specifications differ in what is preserved beyond DARA. Multiplicative risk yields a shifter-weight decomposition and preserves CRRA form; additive income risk—the relevant case for dynastic precautionary savings—preserves DARA but not functional form.

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1 Introduction

Warm-glow bequest motives are a standard device in quantitative macroeconomics and structural models of life-cycle consumption-savings decisions. A parent is assumed to value bequests directly, and the descendant’s problem need not be modelled in detail. Following Abel and Warshawsky (1988), who showed that altruism under certainty can be reduced to this form, a large applied literature has adopted warm-glow preferences with decreasing absolute risk aversion—often constant relative risk aversion (CRRA) or a shifted power function—and calibrated them to wealth and bequest data.¹

Whether decreasing absolute risk aversion (DARA) is actually implied by an underlying altruistic framework, and what else is implied, depends on the structure of descendant risk. This dependence has received little attention in the literature, despite growing empirical evidence that the dynastic precautionary motive—parents saving to insure children against income risk—is quantitatively important (Boar, 2021).

This paper provides a characterisation. I establish that DARA of the reduced-form bequest motive is inherited from DARA of the descendant’s continuation value under both multiplicative and additive risk, for arbitrary non-parametric preferences. The proof relies on a Cauchy–Schwarz inequality applied to the derivative characterisation of DARA and is self-contained. The result justifies the widespread DARA assumption but is also limited: DARA is a weak restriction, and a natural question is whether the risk structure of the economy provides further discipline.

On this question the two canonical specifications diverge. Multiplicative descendant risk—arising when bequests earn idiosyncratic returns (e.g. Constantinides and Duffie, 1996)—preserves functional form: under pure CRRA period utility, the warm-glow bequest motive is itself CRRA with the same curvature parameter and a constant risk adjustment independent of the bequest level. This is similar to results under certainty derived by Abel and Warshawsky (1988). Additive income risk—the formulation in models following Bewley (1986), İmrohoroglu (1989), Huggett (1993), and Aiyagari (1994) or Deaton (1989), Zeldes (1989), and Carroll (1997)—preserves DARA but not functional form. Thus, the reduced-form preference for bequests differs from the primitive of descendant utility. This delivers a risk adjustment that varies with bequests, and no shifter-weight decomposition obtains. Since additive income risk is the relevant specification for the dynastic precautionary channel documented by Boar (2021) and for the broader class of heterogeneous-agent models in which warm-glow bequests appear, this asymmetry has practical content: in these settings, the researcher’s choice of functional form is not disciplined by the underlying altruism.

2 The descendant’s problem and parental warm-glow

Following Abel and Warshawsky (1988), I derive the parent’s reduced-form bequest motive from a primitive descendant problem. A parent chooses bequest $b \geq 0$ at the end of life; a stochastic state

¹Among many others, De Nardi (2004), Hurd (1989), Kopczuk and Lupton (2007), Ameriks et al. (2011) and De Nardi et al. (2025).

ζ summarises the descendant's permanent income, demographics, and any other payoff-relevant heterogeneity. In general, the descendant's expected continuation utility is a state-dependent value function $V^*(b, \zeta)$. In the special case where future labour income can be summarised by a deterministic present value H_ζ and there are no borrowing-constraint effects that make the value of future income depend on the bequest, V^* collapses to

$$V^*(b, \zeta) = J(b + H_\zeta), \quad (1)$$

where J is the value function of the descendant's life-cycle cake-eating problem over total resources $W = b + H_\zeta$,

$$J(W) = \max_{\{c_t\}_{t=0}^T} \sum_{t=0}^T \beta^t u(c_t) \quad \text{s.t.} \quad \sum_{t=0}^T R^{-t} c_t \leq W, \quad (2)$$

with period utility u strictly increasing, strictly concave, three-times differentiable, prudent, and DARA. In general, the horizon T may be finite, infinite, or stochastic; u may depend on age. Under standard convexity, J inherits four key properties: monotonicity and concavity (Stokey et al., 1989, Ch. 9), differentiability (Benveniste and Scheinkman, 1979), higher-order smoothness (Santos, 1991), and DARA (Epstein, 1983). H_ζ has the natural interpretation of human wealth in the complete-markets or certainty-equivalent benchmark; in the incomplete-markets case the primitive remains $V^*(\cdot, \zeta)$. The analysis below assumes the descendant's indirect utility, conditional on ζ , satisfies DARA. This is treated as a primitive of the problem, but can be established from lower level assumptions on the period utility function.

At the bequest date the parent does not know ζ , hence does not know H_ζ in the specialised example; the descendant, by contrast, observes ζ before consumption choices, consistent with the deterministic problem in (2). Writing $h(b, \zeta)$ for the descendant's eventual total resources, the parent's expected continuation payoff is

$$v(b) = \mathbb{E}[V^*(h(b, \zeta))], \quad (3)$$

a deterministic function of b —the warm-glow representation. Two canonical specifications follow:²

Multiplicative risk. $h(b, Z) = bZ$, with $Z > 0$ and $\mathbb{E}[Z] = 1$. Z captures idiosyncratic returns on the estate (Constantinides and Duffie, 1996) or the compounded effect of permanent shocks to descendant earnings.

Additive risk. $h(b, Y) = b + Y$, with $\mathbb{E}[Y] = 0$ and $b + Y \in \text{dom}(V^*)$ almost surely. Y summarises the parent's residual uncertainty about H_ζ and is the channel through which the dynastic precautionary motive of Boar (2021) operates.

The remainder of the paper studies how the distribution of ζ shapes v .

²Differentiation under the expectation requires $\mathbb{E}[Z^k |V^{*(k)}(bZ)|] < \infty$ in the multiplicative case and $\mathbb{E}[|V^{*(k)}(b + Y)|] < \infty$ in the additive case, for $k \leq 3$. Under CRRA the binding conditions are $\mathbb{E}[Z^{1-\gamma}] < \infty$ (a left-tail restriction on Z when $\gamma > 1$) and $b + Y > 0$ a.s.

3 DARA preservation

The Arrow–Pratt measure of absolute risk aversion is $A_f(x) = -f''(x)/f'(x)$. A function f with $f' > 0$ and $f'' < 0$ exhibits *decreasing absolute risk aversion* (DARA) if A_f is strictly decreasing, which holds if and only if

$$f'''(x) f'(x) > f''(x)^2 \quad (4)$$

for all x in the domain.

Theorem 1. *Let V^* be DARA. Then v defined by (3) is DARA under both multiplicative and additive risk. Strict DARA of v holds whenever V^* is not constant absolute risk aversion (CARA). When V^* is CARA, v remains CARA under additive risk but becomes strictly DARA under multiplicative risk with non-degenerate Z .*

Proof. The proof chains two inequalities: a pointwise bound from DARA of V^* and a Cauchy–Schwarz inequality on expectations.

Additive case. With $h = b + Y$, $v^{(k)}(b) = \mathbb{E}[V^{*(k)}(b + Y)]$ for $k = 1, 2, 3$. DARA of v requires

$$\mathbb{E}[V^{*'''}(b + Y)] \mathbb{E}[V^{*'}(b + Y)] \geq (\mathbb{E}[V^{*''}(b + Y)])^2. \quad (5)$$

DARA of V^* gives $V^{*''}(x)^2 \leq V^{*'''}(x) V^{*'}(x)$ pointwise. Since $V^{*''} < 0$, this yields $-V^{*''}(x) \leq \sqrt{V^{*'''}(x) V^{*'}(x)}$. Taking expectations and applying Cauchy–Schwarz to $\sqrt{V^{*'''}(b + Y)}$ and $\sqrt{V^{*'}(b + Y)}$:

$$(\mathbb{E}[-V^{*''}])^2 \leq \left(\mathbb{E}[\sqrt{V^{*'''} V^{*'}}] \right)^2 \leq \mathbb{E}[V^{*'''}] \mathbb{E}[V^{*'}],$$

which is (5).

Multiplicative case. With $h = bZ$, $v^{(k)}(b) = \mathbb{E}[Z^k V^{*(k)}(bZ)]$. From pointwise DARA, $Z^2(-V^{*''}(bZ)) \leq Z^2 \sqrt{V^{*'''}(bZ) V^{*'}(bZ)}$. Cauchy–Schwarz applied to $Z^{3/2} \sqrt{V^{*'''}(bZ)}$ and $Z^{1/2} \sqrt{V^{*'}(bZ)}$ gives

$$(\mathbb{E}[Z^2(-V^{*''})])^2 \leq \mathbb{E}[Z^3 V^{*'''}] \mathbb{E}[Z V^{*'}].$$

Strictness. An overall strict inequality requires at least one of the two steps to be strict. The pointwise step is strict whenever V^* is not constant absolute risk aversion (CARA), which suffices for strict DARA of v under either risk structure regardless of the shock distribution. When V^* is CARA, the pointwise step holds with equality. Under additive risk, the Cauchy–Schwarz step also holds with equality (since $V^{*''}/V^{*'}$ is then constant), so v is itself CARA. Under multiplicative risk with non-degenerate Z , the Cauchy–Schwarz proportionality condition fails, giving strict inequality and hence strict DARA of v . $\square$

The result imposes no parametric restriction on V^* ; the iso-elastic assumption in Abel and Warshawsky (1988) is not needed for DARA inheritance. The economic intuition is that $v'(b)$ is an average of marginal utilities across unresolved descendant states. Under DARA, marginal

utility is a convex function whose curvature decreases with resources. Averaging across states preserves this structure because the Cauchy–Schwarz inequality disciplines how the dispersion of marginal utilities interacts with their level—the same mechanism that underlies the preservation of log-concavity under convolution.

Remark 1. Theorem 1 can be viewed as an inheritance result for the expectation step of a dynamic-programming operator. Carroll and Kimball (1996) establish the parallel inheritance of concavity within the Hyperbolic Absolute Risk Aversion (HARA) class—Toda (2021) shows HARA is necessary—whereas the logic of this proof can be used to show DARA is preserved under no parametric restriction.

4 Precautionary bequests

Having established that risk-aversion properties carry over from V^* to v , I now ask how increased descendant risk shifts the marginal value of bequests. A mean-preserving spread in the descendant’s shock raises $v'(b)$ under standard conditions.³

Proposition 1. *Under additive risk, if $V^{*'''} > 0$ (prudence; Kimball, 1990), a mean-preserving spread in Y raises $v'(b) = \mathbb{E}[V^{*'}(b + Y)]$.*

Proof. $V^{*'}$ is convex under prudence. Apply Jensen’s inequality. $\square$

Proposition 2. *Under multiplicative risk, if $\phi(z) = zV^{*'}(bz)$ is convex in z —equivalently, if relative prudence $-xV^{*'''}(x)/V^{*''}(x) \geq 2$ —then a mean-preserving spread in Z raises $v'(b) = \mathbb{E}[ZV^{*'}(bZ)]$.*

Proof. $\phi''(z) = b[2V^{*''}(bz) + bzV^{*'''}(bz)] \geq 0$ under the stated condition. $\square$

For CRRA utility with coefficient γ , relative prudence is $\gamma + 1$, so Proposition 2 applies when $\gamma \geq 1$.

Proposition 1 provides the theoretical counterpart of the empirical finding in Boar (2021) that parental consumption responds to children’s permanent income uncertainty. The result requires only prudence of the continuation value, a condition widely maintained in calibrated models.

5 Multiplicative risk: additional structure

Under multiplicative risk, the marginal bequest motive admits the decomposition

$$v'(b) = V^{*'}(b) \mathbb{E} \left[\frac{Z V^{*'}(bZ)}{V^{*'}(b)} \right], \quad (6)$$

³A random variable $\tilde{Y}$ is a *mean-preserving spread* of Y if $\tilde{Y}$ has the same mean as Y and dominates Y in convex order, i.e., $\mathbb{E}[g(\tilde{Y})] \geq \mathbb{E}[g(Y)]$ for every convex g .

separating a certainty-equivalent term from a risk adjustment (see Abel and Warshawsky, 1988, for a related HARA characterisation). When V^* is CRRA, as in many workhorse models, this imposes additional restrictions on the joy-of-giving bequest motive.

Proposition 3. *Under multiplicative risk with $V^*(x) = x^{1-\gamma}/(1-\gamma)$, the reduced form is $v(b) = \mathbb{E}[Z^{1-\gamma}] \cdot b^{1-\gamma}/(1-\gamma)$: CRRA with the same γ and a risk adjustment independent of b .*

Proof. $V^*(bZ) = Z^{1-\gamma} b^{1-\gamma}/(1-\gamma)$. Take expectations. $\square$

The preservation of functional form relies on the homogeneity of pure CRRA utility; it does not extend to shifted power functions $V^*(x) = (x + \kappa)^{1-\gamma}/(1-\gamma)$ with $\kappa > 0$, where the interaction of κ with multiplicative scaling breaks the factorisation. Under shifted CRRA with $\kappa > 0$, the reduced form $v(b) = \mathbb{E}[(bZ + \kappa)^{1-\gamma}]/(1-\gamma)$ remains DARA by Theorem 1 but is no longer a member of the HARA class with the same parameters.

6 Additive risk: DARA without further structure

Under additive income risk, DARA is preserved and precautionary motives arise, but the additional structures that multiplicative risk delivers are absent.

Importantly, the functional form of descendant utility is not preserved. Under CRRA, $v(b) = \mathbb{E}[(b + Y)^{1-\gamma}]/(1-\gamma)$,⁴ which differs from $b^{1-\gamma}/(1-\gamma)$ whenever Y is non-degenerate. There is no factorisation of $v'(b) = \mathbb{E}[V^{*'}(b + Y)]$ that separates the bequest level from the distribution of Y : the shock enters inside $V^{*'}$, not through scaling. The ratio $v'(b)/V^{*'}(b)$ depends on b and converges to one as the bequest grows large relative to the support of Y .

This is the empirically prevalent case. Additive income risk is the specification in the Bewley–Imrohoroglu–Huggett–Aiyagari tradition, in the dynastic precautionary savings framework of Boar (2021), and in most quantitative life-cycle models with bequest motives. In all such settings, assuming DARA for the reduced-form bequest motive is justified by Theorem 1. Assuming a specific functional form—CRRA, shifted CRRA, or otherwise—is a separate modeling choice that the underlying altruism does not discipline.

What does this mean in practice? The true reduced form under additive risk, $v(b) = \mathbb{E}[(b + Y)^{1-\gamma}]/(1-\gamma)$, has curvature that varies with b : absolute risk aversion is high at small bequests, where the additive shock is large relative to resources, and declines toward the CRRA benchmark as bequests grow. This pattern is naturally captured by the Stone–Geary or shifted-CRRA specifications commonly adopted in applied work—for instance, the $(b + \kappa)^{1-\gamma}/(1-\gamma)$ form in De Nardi (2004)—which generate declining curvature through a shift parameter κ . The connection is intuitive: κ plays a role analogous to the dispersion of Y , absorbing the effect of descendant risk into a single reduced-form parameter. The approximation is reasonable but necessarily incomplete, since the dependence of curvature on b in $\mathbb{E}[(b + Y)^{1-\gamma}]$ is shaped by the full distribution of Y , including its higher moments, in a way that a single parameter cannot fully represent. This offers a useful lens

⁴Well-defined under the support condition $b + Y > 0$ a.s. introduced in footnote 2.

on cross-study variation in calibrated values of κ : such differences may reflect not only differences in the strength of the bequest motive but also differences in the descendant risk environment being approximated.

7 Concluding remarks

The practice of assuming DARA warm-glow bequests is consistent with the underlying altruism whenever the descendant’s continuation value is itself smooth and DARA—conditions widely maintained in calibrated models. What the altruism does not supply is the functional form. Under multiplicative descendant risk, functional form is pinned down: pure CRRA continuation utility implies CRRA bequests. Under additive income risk—the dominant specification in quantitative work—it is not. The functional-form assumption carries independent empirical content and should be assessed against data on bequests and wealth, not inherited from the primitive by appeal to the altruistic motive.

In this sense, reduced-form bequest preferences are ultimately estimated terminal value functions (e.g., Keane and Wolpin, 1994, 1997). Consequently, differences in calibrated bequest-motive parameters across studies may reflect not only differences in the strength of altruism but also differences in the descendant risk environment—or simply in the approximation error from imposing a particular functional form. The results here suggest that care is needed when mapping estimated or calibrated parameters to an underlying altruistic structure.

A Declaration of generative AI and AI-assisted technologies in the writing process.

During the preparation of this work the author used ChatGPT and Claude in order to copy edit the written text in the paper and proofread mathematical content. After using these services the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

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